# Adaptive GA Deep Feature Selector for Lung Histopathological Image Classification

## Experiment Results Report

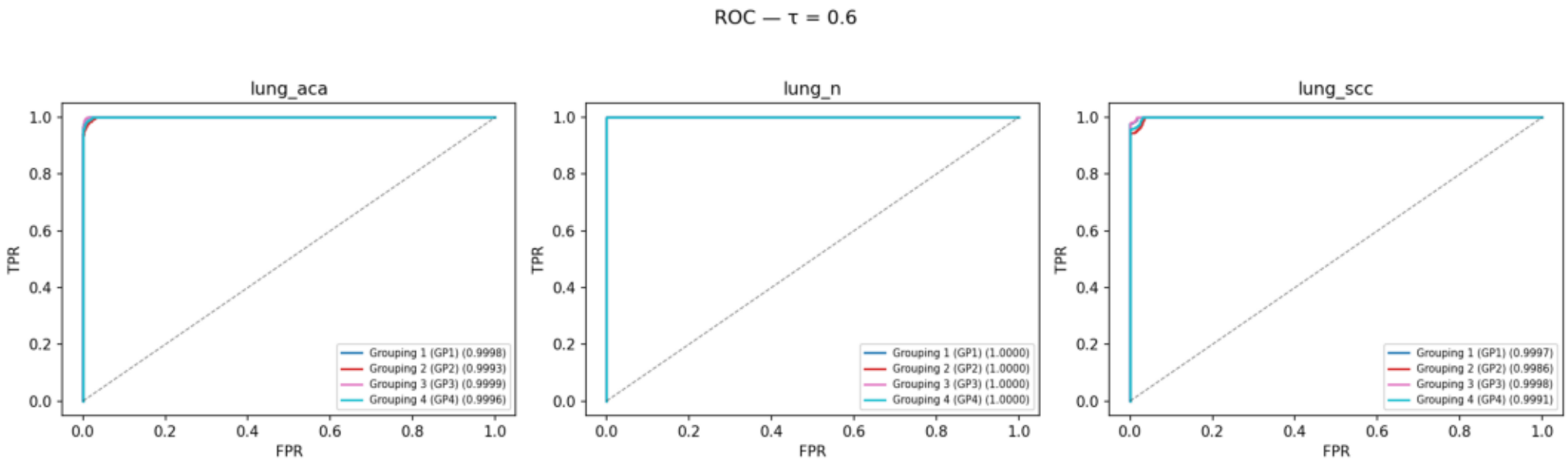
DenseNet121 + SE Attention + Adaptive GA + GP Ensemble Fusion
Dataset: LC25000 (15,000 images, 3 classes)

Anshul Saxena, Advik Kashi Vishwanath, Kritii Gupta, Chandrima Debnath
Supervisor: Prof. Swati Hait, BITS Pilani Hyderabad

## Table 1: Effect of Saturation Threshold τ = 0.6 on Grouping Operators

| Method | Acc | Prec | Sens | Spec | F1 | AUC | MCC |
|---|---|---|---|---|---|---|---|
| Grouping 1 (GP1) | 0.9903 | 0.9906 | 0.9903 | 0.9952 | 0.9904 | 0.9998 | 0.9856 |
| Grouping 2 (GP2) | 0.9783 | 0.9797 | 0.9783 | 0.9892 | 0.9784 | 0.9993 | 0.9681 |
| **Grouping 3 (GP3)** | **0.9923** | **0.9925** | **0.9923** | **0.9962** | **0.9923** | **0.9999** | **0.9886** |
| Grouping 4 (GP4) | 0.984 | 0.9847 | 0.984 | 0.992 | 0.984 | 0.9996 | 0.9763 |

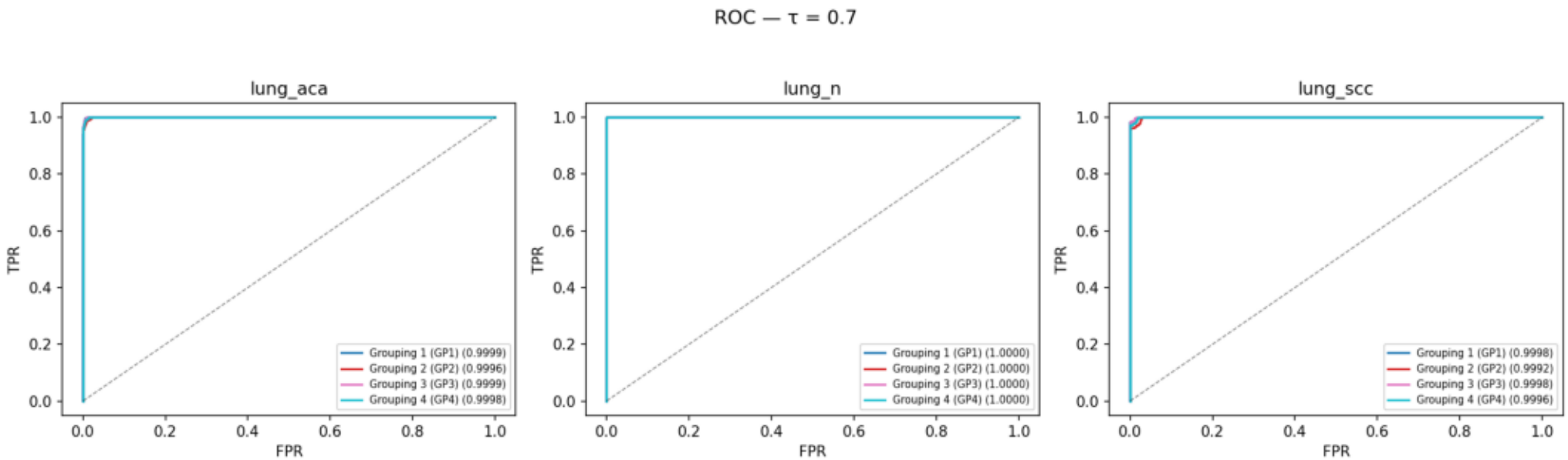
# ROC Curves — τ = 0.6



## Table 1: Effect of Saturation Threshold τ = 0.7 on Grouping Operators

| Method | Acc | Prec | Sens | Spec | F1 | AUC | MCC |
|---|---|---|---|---|---|---|---|
| Grouping 1 (GP1) | 0.9923 | 0.9925 | 0.9923 | 0.9962 | 0.9923 | 0.9999 | 0.9886 |
| Grouping 2 (GP2) | 0.9853 | 0.986 | 0.9853 | 0.9927 | 0.9854 | 0.9996 | 0.9783 |
| **Grouping 3 (GP3)** | **0.9933** | **0.9935** | **0.9933** | **0.9967** | **0.9933** | **0.9999** | **0.9901** |
| Grouping 4 (GP4) | 0.989 | 0.9894 | 0.989 | 0.9945 | 0.989 | 0.9998 | 0.9837 |

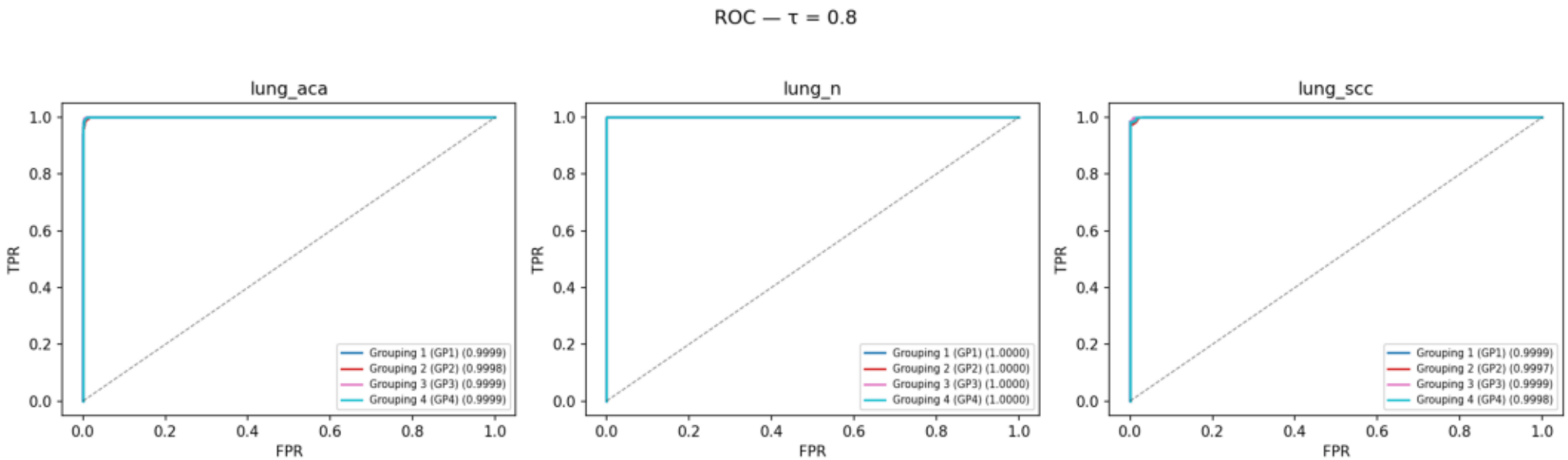
# ROC Curves — τ = 0.7



## Table 1: Effect of Saturation Threshold τ = 0.8 on Grouping Operators

| Method | Acc | Prec | Sens | Spec | F1 | AUC | MCC |
|---|---|---|---|---|---|---|---|
| Grouping 1 (GP1) | 0.9937 | 0.9938 | 0.9937 | 0.9968 | 0.9937 | 0.9999 | 0.9905 |
| Grouping 2 (GP2) | 0.9897 | 0.99 | 0.9897 | 0.9948 | 0.9897 | 0.9998 | 0.9846 |
| **Grouping 3 (GP3)** | **0.995** | **0.9951** | **0.995** | **0.9975** | **0.995** | **0.9999** | **0.9925** |
| Grouping 4 (GP4) | 0.9927 | 0.9928 | 0.9927 | 0.9963 | 0.9927 | 0.9999 | 0.9891 |

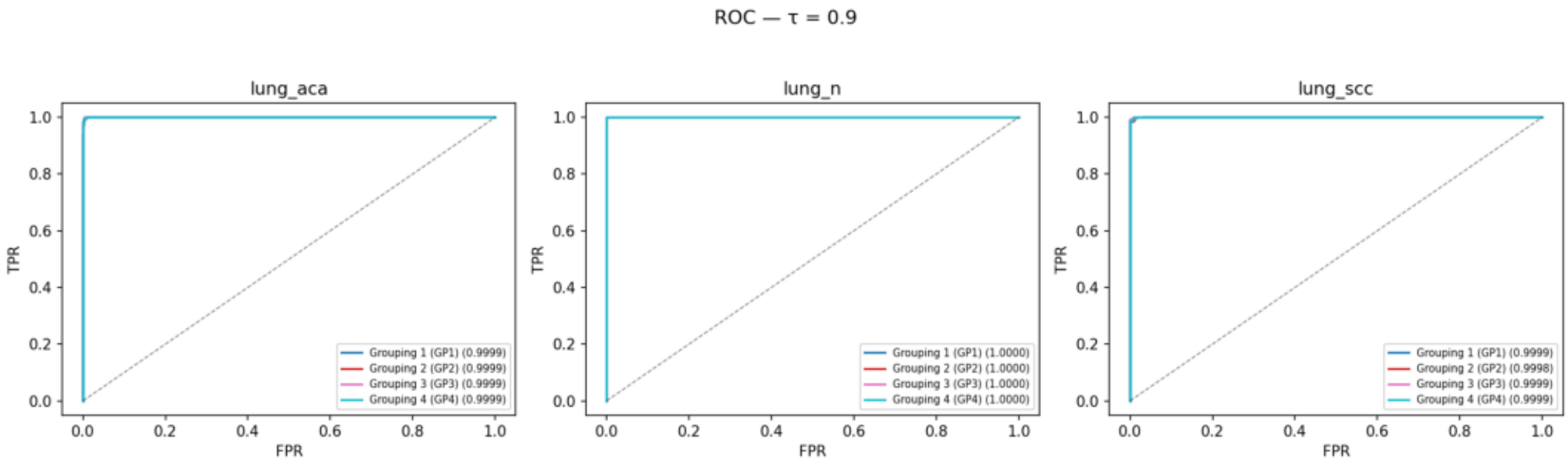
# ROC Curves — τ = 0.8



## Table 1: Effect of Saturation Threshold τ = 0.9 on Grouping Operators

| Method | Acc | Prec | Sens | Spec | F1 | AUC | MCC |
|---|---|---|---|---|---|---|---|
| Grouping 1 (GP1) | 0.996 | 0.996 | 0.996 | 0.998 | 0.996 | 0.9999 | 0.994 |
| Grouping 2 (GP2) | 0.993 | 0.9931 | 0.993 | 0.9965 | 0.993 | 0.9999 | 0.9896 |
| **Grouping 3 (GP3)** | **0.997** | **0.997** | **0.997** | **0.9985** | **0.997** | **1.0** | **0.9955** |
| Grouping 4 (GP4) | 0.994 | 0.9941 | 0.994 | 0.997 | 0.994 | 0.9999 | 0.991 |

# ROC Curves — τ = 0.9



# Table 1 (Combined): All τ Values — Grouping Operator Performance

| τ | Method | Acc | Prec | Sens | Spec | F1 | AUC | MCC |
|---|---|---|---|---|---|---|---|---|
| 0.6 | Grouping 1 (GP1) | 0.9903 | 0.9906 | 0.9903 | 0.9952 | 0.9904 | 0.9998 | 0.9856 |
| 0.6 | Grouping 2 (GP2) | 0.9783 | 0.9797 | 0.9783 | 0.9892 | 0.9784 | 0.9993 | 0.9681 |
| 0.6 | Grouping 3 (GP3) | 0.9923 | 0.9925 | 0.9923 | 0.9962 | 0.9923 | 0.9999 | 0.9886 |
| 0.6 | Grouping 4 (GP4) | 0.984 | 0.9847 | 0.984 | 0.992 | 0.984 | 0.9996 | 0.9763 |
| 0.7 | Grouping 1 (GP1) | 0.9923 | 0.9925 | 0.9923 | 0.9962 | 0.9923 | 0.9999 | 0.9886 |
| 0.7 | Grouping 2 (GP2) | 0.9853 | 0.986 | 0.9853 | 0.9927 | 0.9854 | 0.9996 | 0.9783 |
| 0.7 | Grouping 3 (GP3) | 0.9933 | 0.9935 | 0.9933 | 0.9967 | 0.9933 | 0.9999 | 0.9901 |
| 0.7 | Grouping 4 (GP4) | 0.989 | 0.9894 | 0.989 | 0.9945 | 0.989 | 0.9998 | 0.9837 |
| 0.8 | Grouping 1 (GP1) | 0.9937 | 0.9938 | 0.9937 | 0.9968 | 0.9937 | 0.9999 | 0.9905 |
| 0.8 | Grouping 2 (GP2) | 0.9897 | 0.99 | 0.9897 | 0.9948 | 0.9897 | 0.9998 | 0.9846 |
| 0.8 | Grouping 3 (GP3) | 0.995 | 0.9951 | 0.995 | 0.9975 | 0.995 | 0.9999 | 0.9925 |
| 0.8 | Grouping 4 (GP4) | 0.9927 | 0.9928 | 0.9927 | 0.9963 | 0.9927 | 0.9999 | 0.9891 |
| 0.9 | Grouping 1 (GP1) | 0.996 | 0.996 | 0.996 | 0.998 | 0.996 | 0.9999 | 0.994 |
| 0.9 | Grouping 2 (GP2) | 0.993 | 0.9931 | 0.993 | 0.9965 | 0.993 | 0.9999 | 0.9896 |
| **0.9** | **Grouping 3 (GP3)** | **0.997** | **0.997** | **0.997** | **0.9985** | **0.997** | **1.0** | **0.9955** |
| 0.9 | Grouping 4 (GP4) | 0.994 | 0.9941 | 0.994 | 0.997 | 0.994 | 0.9999 | 0.991 |

# Table 2: Best Grouping Operator vs Individual Classifiers

| Method | Acc | Prec | Sens | Spec | F1 | AUC | MCC |
|---|---|---|---|---|---|---|---|
| **st Grouping (Grouping 3 (GP3) τ=** | **0.997** | **0.997** | **0.997** | **0.9985** | **0.997** | **1.0** | **0.9955** |
| KNN (Single) | 0.9957 | 0.9957 | 0.9957 | 0.9978 | 0.9957 | 0.9997 | 0.9935 |
| SVM (Single) | 0.9777 | 0.9778 | 0.9777 | 0.9888 | 0.9777 | 0.9983 | 0.9666 |
| RF (Single) | 0.984 | 0.984 | 0.984 | 0.992 | 0.984 | 0.9991 | 0.976 |
| LR (Single) | 0.9777 | 0.9777 | 0.9777 | 0.9888 | 0.9777 | 0.998 | 0.9665 |
| XGBoost (Single) | 0.9927 | 0.9927 | 0.9927 | 0.9963 | 0.9927 | 0.9997 | 0.989 |

## Table 3: Comparison with State-of-the-Art Methods on LC25000

| Method | Accuracy | F1 | Sensitivity |
|---|---|---|---|
| Masud et al. (2021) | 0.9633 | 0.963 | 0.9633 |
| Mangal et al. (2020) | 0.9789 | 0.9785 | 0.9789 |
| Nishio et al. (2021) | 0.95 | 0.949 | 0.95 |
| Hatuwal & Thapa (2020) | 0.972 | 0.9715 | 0.972 |
| Talukder et al. (2022) | 0.981 | 0.9808 | 0.981 |
| Hage Chehade et al. (2022) | 0.9625 | 0.962 | 0.9625 |
| **Proposed Method (Ours)** | **0.9957** | **0.9957** | **0.9957** |

## Table 4: Effect of Number of GA Generations

| Gens | Feat | Acc | Prec | Sens | Spec | F1 | AUC | MCC |
|---|---|---|---|---|---|---|---|---|
| 50 | 521 | 0.9957 | 0.9957 | 0.9957 | 0.9978 | 0.9957 | 0.9997 | 0.9935 |
| **100** | **505** | **0.9967** | **0.9967** | **0.9967** | **0.9983** | **0.9967** | **1.0** | **0.995** |
| 150 | 482 | 0.9967 | 0.9967 | 0.9967 | 0.9983 | 0.9967 | 1.0 | 0.995 |

## Table 5: Effect of Crossover Probability

| CX | Feat | Acc | Prec | Sens | Spec | F1 | AUC | MCC |
|---|---|---|---|---|---|---|---|---|
| 0.4 | 521 | 0.9957 | 0.9957 | 0.9957 | 0.9978 | 0.9957 | 0.9997 | 0.9935 |
| **0.5** | **509** | **0.996** | **0.996** | **0.996** | **0.998** | **0.996** | **1.0** | **0.994** |
| 0.6 | 514 | 0.9957 | 0.9957 | 0.9957 | 0.9978 | 0.9957 | 1.0 | 0.9935 |

# Table 6: Genetic Algorithm Hyperparameters

| Parameter | Value |
|---|---|
| Population Size | 40 |
| Generations | 50 |
| Crossover Probability | 0.8 |
| Mutation Probability | 0.1 |
| Bit-flip Probability | 0.05 |
| Inner CV Folds | 3 |
| Classifier (fitness) | KNN (k=5, distance) |
| L0 Penalty | 0.001 |
| Random Seed | 42 |